

TECHNISCHE UNIVERSITÄT DRESDEN

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INSTITUTE OF SYSTEMS ARCHITECTURE
CHAIR OF DATABASES
PROF. DR.-ING. WOLFGANG LEHNER

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Active Cache-Aware Hardware Adaptation: Cache Engine for Trie-Based Index Structures

Benjamin-Elias Probst
(Born 11 April 1996 in Potsdam)

Professor: apl. Prof. Dr.-Ing. Dirk Habich

Tutor: apl. Prof. Dr.-Ing. Dirk Habich, Dr.-Ing. Alexander Krause

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Hier Aufgabenstellung einfügen!

Selbstständigkeitserklärung

Hiermit erkläre ich, dass ich die von mir am heutigen Tag dem Prüfungsausschuss der Fakultät Informatik eingereichte Diplomarbeit zum Thema:

*Active Cache-Aware Hardware Adaptation:
Cache Engine for Trie-Based Index Structures*

vollkommen selbstständig verfasst und keine anderen als die angegebenen Quellen und Hilfsmittel benutzt sowie Zitate kenntlich gemacht habe.

Dresden, den 1. Juni 2026

Benjamin-Elias Probst

Kurzfassung

Platzhalter – deutsche Kurzfassung folgt.

Abstract

Placeholder – English abstract to follow.

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1 Introduction

1.1 Motivation

Modern in-memory databases rely on cache-aware index structures such as the Adaptive Radix Tree [LKN13].

1.2 Problem Statement

1.3 Research Questions

RQ1 (Composition)

RQ2 (Measurement methodology)

RQ3 (Probe comparison)

1.4 Objectives and Contributions

1.5 Structure of the Thesis

2 Fundamentals

2.1 Cache Hierarchy and Cache Awareness

2.2 Trie-Based Index Structures

2.3 The Unified Comparison Interface (`std::map`)

2.4 YCSB Workloads

2.5 C++23 Metaprogramming

3 State of the Art

3.1 Trie Family (Cluster A)

3.2 Hybrid and B⁺ Family (Cluster B)

3.3 Layout Theory (Cluster C)

3.4 Prefetching (Clusters D and E)

3.5 Synchronization (Cluster F)

3.6 Allocator Research (A01–A23)

3.7 Research Gap

4 Concept and Architecture

4.1 The Axis Library Framework

4.2 Four-Subsystem Separation (M Model)

4.3 Three-Layer Hierarchy

4.4 ABI-Stable C++23 Module Interface

4.5 The Building-Block Axes

4.6 PRT-ART as Probe

4.7 CacheEngineBuilder and Three-Stage Evaluation

5 Implementation

5.1 Three-Repository Architecture

5.2 Adapters for State-of-the-Art Algorithms and Allocators

5.3 Permutation Code Generation and Flag System

5.4 Concept/C RTP Realization of the Axes

5.5 Concurrency and Reclamation

5.6 Telemetry against Cache-Coherence Effects

5.7 Measurement Pipeline

6 Evaluation Methodology

6.1 Three Measurement Series

6.2 ExperimentDriver Phases

6.3 Hypotheses

6.4 Workloads and Datasets

6.5 Experimental Platforms

6.6 Permutation Explosion and Reduction

7 Results and Evaluation

7.1 Evaluation Pipeline

7.2 Series A: PRT-ART vs. State of the Art

7.3 Series B: Cache-Engine Permutations

7.4 Series C: Full Join

7.5 Axis Sensitivity Analysis

7.6 Discussion

8 Conclusion and Outlook

8.1 Answering the Research Questions

8.2 Limitations

8.3 Outlook

Bibliography

- [LKN13] Viktor Leis, Alfons Kemper, and Thomas Neumann. The adaptive radix tree: ARTful indexing for main-memory databases. In *29th IEEE International Conference on Data Engineering (ICDE)*, pages 38–49, 2013.

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A Complete Measurement Series

B Code Structure

C Glossary

D Building-Block Matrix

E Architecture Decisions

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